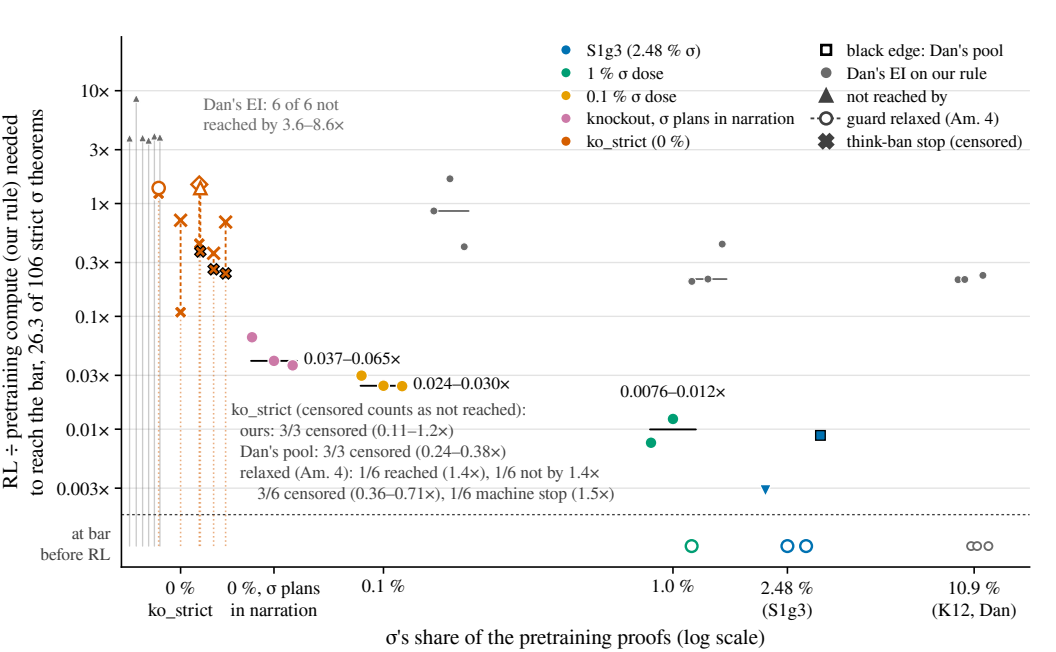

One marker per seed. Bar: 26.3 of 106 held-out theorems that need classical proof by contradiction (σ, Dan's strict set), as E32 (the expected number a 32-sample draw solves), the median of our three S1g3 bases before RL. y: (6 × trained + 2 × sampled tokens) ÷ (6 × Stage-1 tokens), prompt prefill excluded. Filled: first crossing, interpolated in log y between checkpoints; ▾: at the first checkpoint (an upper bound); hollow: at the bar before RL; ▲ on a thin line: not reached by that y; cross on a dotted line: the recipe's think-ban guard ended the line there (censored; counted as not reached by that y, the conservative reading, since excluding it would flatter the arm); hollow on a dashed line: the guard-relaxed continuation (Amendment 4, secondary): circle reached, triangle not by the cap, x stopped again by the relaxed guard, diamond stopped for machine availability. Colour: GRPO with programmatic thinking traces on our pool (39.5–44.5 % of targets σ-covered); black edge: the same GRPO on Dan's 4,495 EI targets (2.7 %); grey: Dan's T1 expert-iteration ladders on this y (his own x adds re-read state prefill and replay: 1.5–1.9 × these values after round 1; round 1, sampling only, is drawn on the floor). Differences from Dan: GRPO, not EI; a thinking model (programmatic search traces); no pretraining replay; S1g3 bases (σ 2.48 %, not 10.9 %); doses of 0.1 % and 1.0 % of the proofs; ko has no σ proofs, but 0.41 % of its records plan the σ detour in their narration at the effort level the bases trained on; ko_strict removes the 949 records with such a plan at any level.